

Pseudocode

1. Open the file input.txt for reading.
2. Print an error message and exit if the file is not found.
3. Read the entire line from the file into requirements.
4. If the requirements string is too short (less than 2 characters), print an error message and exit.
5. Set shift_dir to the first character of requirements.
6. If shift_dir is not 'l' (left) or 'r' (right), print an error message and exit.
7. Extract the shift value by parsing the digits after the direction character in requirements.
8. Print an error message and exit if no valid shift value is found.
9. The remaining part of the requirements is the input_str to be processed.
10. Set shift_val to $\text{shift_val} \% \text{ALPHABET_LENGTH}$ to ensure it's within the range of 0 to 25.
11. For each character:
 - a. Convert the character to its corresponding number using number_from_letter().
 - b. Depending on the shift_dir, shift the character's number left or right by shift_val.
 - c. If the new number is outside the valid range (negative or greater than 25), adjust it to wrap around within the alphabet.
 - d. Convert the new number back to a character using letter_from_number().
 - e. Append the new character to output_string.
12. Print the output_string containing the transformed characters.